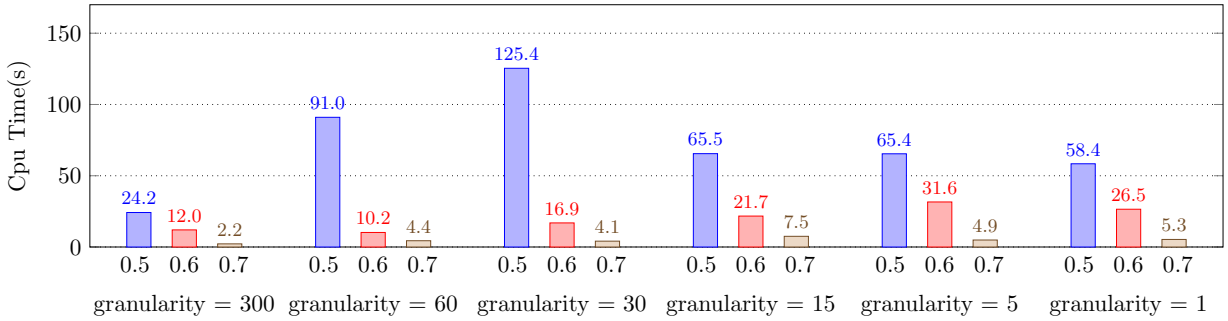
Computational Results for Instance Set S_B

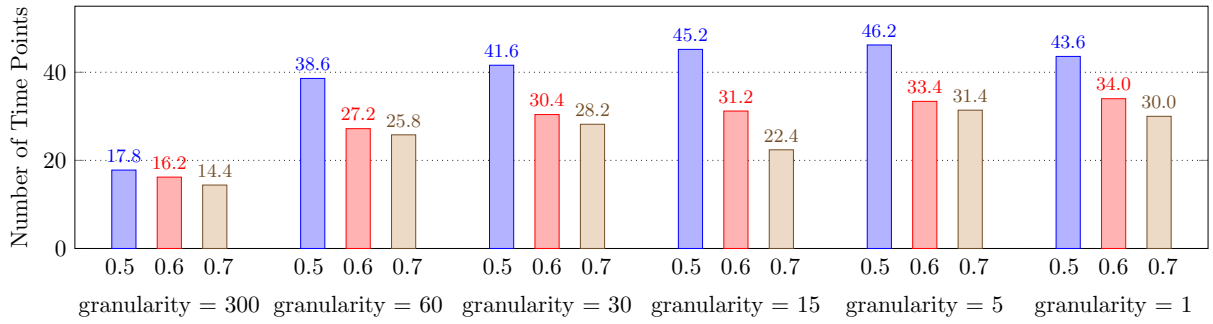
1. Instances with Various Time Granularities

1.1 Uniform weights

The following figures using uniform-weight instances with various time granularities :



(a) Average Computation Time



(b) Average Number of Discovered Time-Points

Figure 1 The Computation Time and Discovered Time-Points of the Obtained Solutions for Uniform-Weight Instances with Different Time Granularities

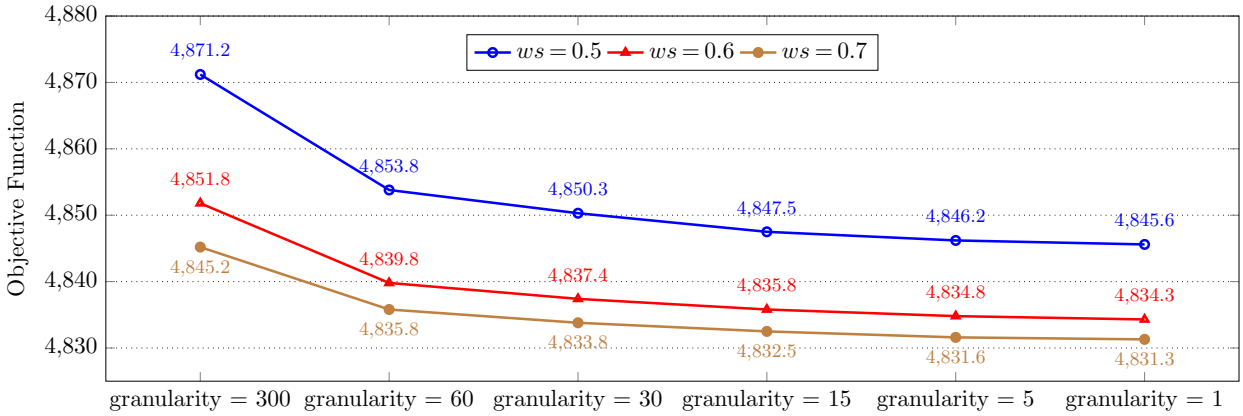


Figure 2 The Average Optimal Objective Value (in 5 Minutes) for Uniform-Weight Instances with Different Granularities

Computational Results for Instance Set S_B

1. Instances with Various Time Granularities

1.1 Uniform weights

Table 1 Results of Medium-Size Uniform-Weight Instances in S_B with Granularity 300

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	4931	0.00	86.67	90-30-1	4909	0.00	27.23	90-30-1	4899	0.00	3.59
90-30-2	4840	0.00	4.86	90-30-2	4806	0.00	1.88	90-30-2	4796	0.00	2.17
90-30-3	4991	0.00	23.90	90-30-3	4978	0.00	23.70	90-30-3	4971	0.00	3.94
90-30-4	4694	0.00	2.52	90-30-4	4679	0.00	4.92	90-30-4	4675	0.00	0.69
90-30-5	4900	0.00	3.44	90-30-5	4887	0.00	2.48	90-30-5	4885	0.00	0.69

Table 2 Results of Medium-Size Uniform-Weight Instances in S_B with Granularity 60

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	4911.00	0.00	366.91	90-30-1	4896.20	0.00	33.61	90-30-1	4891.00	0.00	10.52
90-30-2	4816.20	0.00	41.46	90-30-2	4792.80	0.00	3.07	90-30-2	4785.00	0.00	4.71
90-30-3	4973.00	0.00	26.02	90-30-3	4962.80	0.00	3.28	90-30-3	4961.20	0.00	3.61
90-30-4	4682.40	0.00	14.78	90-30-4	4669.60	0.00	6.98	90-30-4	4666.40	0.00	1.82
90-30-5	4886.80	0.00	5.85	90-30-5	4878.00	0.00	4.17	90-30-5	4875.80	0.00	1.69

Computational Results for Instance Set S_B

1. Instances with Various Time Granularities

1.1 Uniform weights

Table 3 Results of Medium-Size Uniform-Weight Instances in S_B with Granularity 30

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	4907.90	0.00	414.86	90-30-1	4892.60	0.00	34.34	90-30-1	4889.20	0.00	7.42
90-30-2	4810.50	0.00	123.32	90-30-2	4790.30	0.00	4.07	90-30-2	4782.90	0.00	5.39
90-30-3	4970.60	0.00	72.63	90-30-3	4960.50	0.00	3.44	90-30-3	4959.30	0.00	4.74
90-30-4	4678.40	0.00	10.49	90-30-4	4668.10	0.00	38.40	90-30-4	4664.10	0.00	1.73
90-30-5	4884.40	0.00	5.89	90-30-5	4875.70	0.00	4.39	90-30-5	4873.90	0.00	1.66

Table 4 Results of Medium-Size Uniform-Weight Instances in S_B with Granularity 15

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	4904.65	0.00	230.13	90-30-1	4890.90	0.00	55.55	90-30-1	4888.00	0.00	9.78
90-30-2	4805.95	0.00	12.37	90-30-2	4788.10	0.00	4.23	90-30-2	4781.20	0.00	4.59
90-30-3	4967.80	0.00	66.07	90-30-3	4958.55	0.00	4.39	90-30-3	4957.70	0.00	6.46
90-30-4	4677.05	0.00	12.91	90-30-4	4666.95	0.00	40.48	90-30-4	4662.95	0.00	15.50
90-30-5	4882.45	0.00	6.09	90-30-5	4874.55	0.00	4.09	90-30-5	4872.80	0.00	1.54

Table 5 Results of Medium-Size Uniform-Weight Instances in S_B with Granularity 5

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	4903.18	0.00	196.61	90-30-1	4889.90	0.00	110.06	90-30-1	4887.28	0.00	8.18
90-30-2	4804.40	0.00	30.49	90-30-2	4787.08	0.00	3.13	90-30-2	4780.33	0.00	6.50
90-30-3	4966.33	0.00	76.74	90-30-3	4957.42	0.00	3.94	90-30-3	4956.67	0.00	6.65
90-30-4	4675.62	0.00	17.04	90-30-4	4666.03	0.00	37.17	90-30-4	4662.25	0.00	1.81
90-30-5	4881.50	0.00	6.13	90-30-5	4873.57	0.00	4.18	90-30-5	4871.90	0.00	1.59

Table 6 Results of Medium-Size Uniform-Weight Instances in S_B with Granularity 1

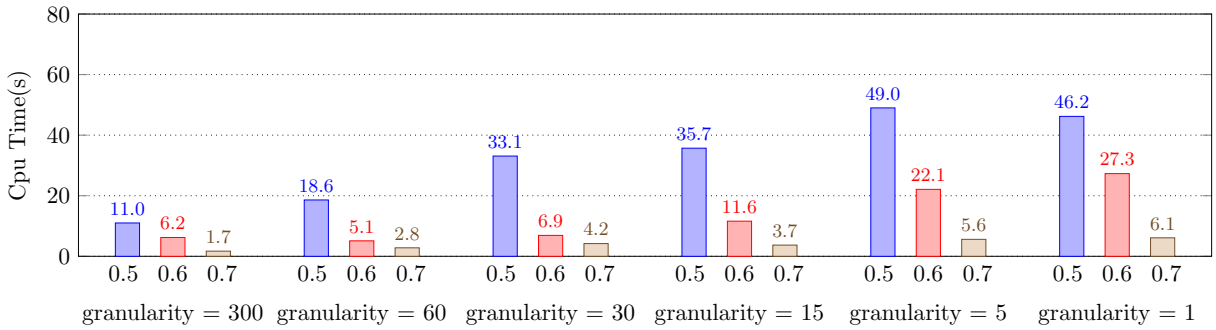
ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	4902.59	0.00	178.31	90-30-1	4889.46	0.00	74.12	90-30-1	4886.93	0.00	10.19
90-30-2	4803.78	0.00	34.45	90-30-2	4786.64	0.00	4.92	90-30-2	4780.06	0.00	4.37
90-30-3	4965.78	0.00	55.23	90-30-3	4956.97	0.00	4.16	90-30-3	4956.28	0.00	9.21
90-30-4	4675.14	0.00	15.25	90-30-4	4665.67	0.00	43.73	90-30-4	4662.03	0.00	1.71
90-30-5	4881.07	0.00	8.85	90-30-5	4873.20	0.00	5.70	90-30-5	4871.57	0.00	1.48

Computational Results for Instance Set S_B

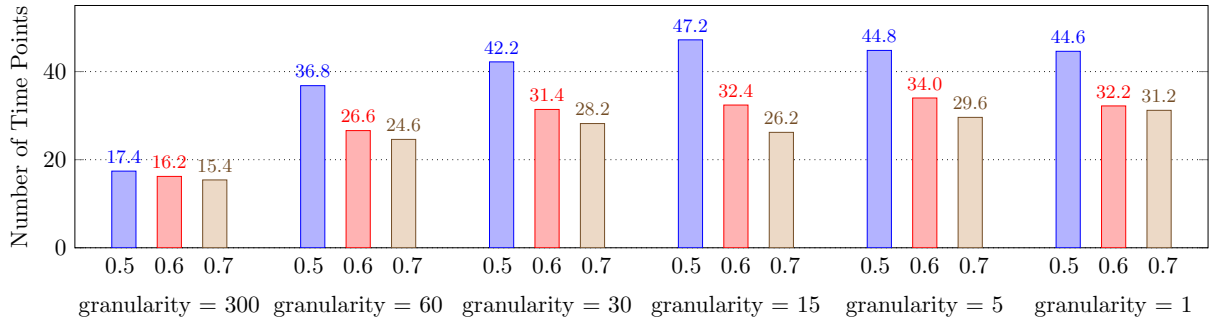
1. Instances with Various Time Granularities

1.2 Standard weights

The following figures using standard-weight instances with various time granularities :



(a) Average Computation Time



(b) Average Number of Discovered Time-Points

Figure 3 The Computation Time and Discovered Time-Points of the Obtained Solutions for Standard-Weight Instances with Different Time Granularities

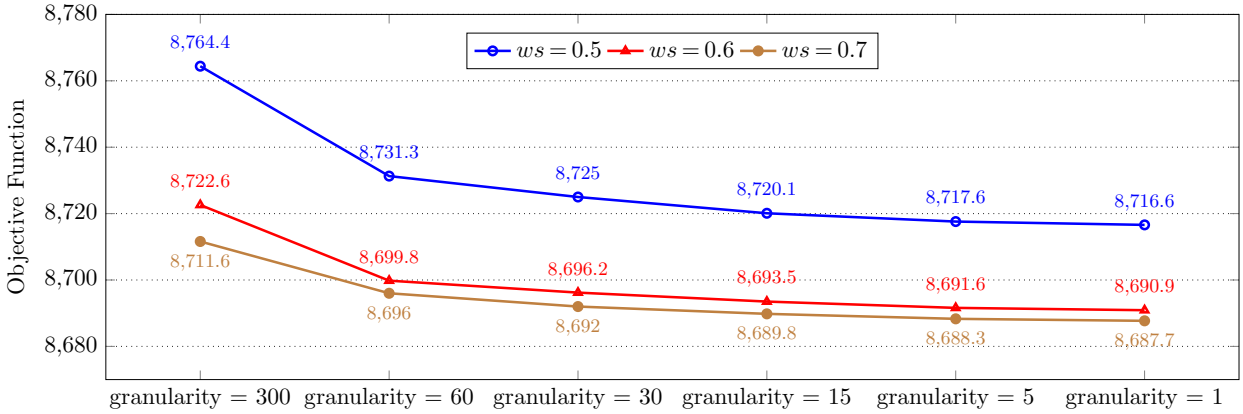


Figure 4 The Average Optimal Objective Value (in 5 Minutes) for Standard-Weight Instances with Different Granularities

Computational Results for Instance Set S_B

1. Instances with Various Time Granularities

1.2 Standard weights

Table 7 Results of Medium-Size Standard-Weight Instances in S_B with Granularity 300

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	9588	0.00	32.99	90-30-1	9544	0.00	22.71	90-30-1	9527	0.00	3.17
90-30-2	8591	0.00	2.40	90-30-2	8515	0.00	1.41	90-30-2	8497	0.00	2.36
90-30-3	8673	0.00	15.90	90-30-3	8648	0.00	1.85	90-30-3	8638	0.00	1.98
90-30-4	8981	0.00	1.83	90-30-4	8942	0.00	3.81	90-30-4	8935	0.00	0.43
90-30-5	7989	0.00	1.98	90-30-5	7964	0.00	1.66	90-30-5	7961	0.00	0.68

Table 8 Results of Medium-Size Standard-Weight Instances in S_B with Granularity 60

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	9551.40	0.00	57.56	90-30-1	9520.80	0.00	14.08	90-30-1	9510.00	0.00	6.05
90-30-2	8540.60	0.00	6.54	90-30-2	8490.20	0.00	1.99	90-30-2	8481.40	0.00	4.18
90-30-3	8641.80	0.00	15.66	90-30-3	8622.80	0.00	1.86	90-30-3	8619.40	0.00	2.66
90-30-4	8955.60	0.00	4.88	90-30-4	8919.40	0.00	4.85	90-30-4	8919.40	0.00	0.70
90-30-5	7967.40	0.00	8.64	90-30-5	7946.00	0.00	2.94	90-30-5	7950.00	0.00	0.81

Computational Results for Instance Set S_B

1. Instances with Various Time Granularities

1.2 Standard weights

Table 9 Results of Medium-Size Standard-Weight Instances in S_B with Granularity 30

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	9541.40	0.00	73.53	90-30-1	9517.60	0.00	21.90	90-30-1	9506.30	0.00	10.60
90-30-2	8535.20	0.00	8.18	90-30-2	8485.90	0.00	2.70	90-30-2	8476.60	0.00	4.37
90-30-3	8636.50	0.00	28.95	90-30-3	8618.90	0.00	3.20	90-30-3	8614.80	0.00	4.87
90-30-4	8949.80	0.00	39.71	90-30-4	8915.50	0.00	4.26	90-30-4	8916.40	0.00	0.64
90-30-5	7962.50	0.00	15.32	90-30-5	7943.10	0.00	2.79	90-30-5	7946.00	0.00	0.78

Table 10 Results of Medium-Size Standard-Weight Instances in S_B with Granularity 15

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	9537.85	0.00	137.63	90-30-1	9515.00	0.00	24.89	90-30-1	9503.40	0.00	10.36
90-30-2	8525.65	0.00	10.22	90-30-2	8482.35	0.00	2.70	90-30-2	8474.70	0.00	4.11
90-30-3	8632.30	0.00	21.14	90-30-3	8616.15	0.00	3.02	90-30-3	8612.40	0.00	2.51
90-30-4	8945.45	0.00	6.84	90-30-4	8912.80	0.00	24.40	90-30-4	8914.30	0.00	0.92
90-30-5	7959.45	0.00	2.88	90-30-5	7941.20	0.00	3.38	90-30-5	7944.55	0.00	0.84

Table 11 Results of Medium-Size Standard-Weight Instances in S_B with Granularity 5

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	9535.28	0.00	157.64	90-30-1	9513.35	0.00	95.76	90-30-1	9501.43	0.00	14.98
90-30-2	8522.70	0.00	20.83	90-30-2	8479.85	0.00	2.43	90-30-2	8472.88	0.00	5.08
90-30-3	8629.95	0.00	28.31	90-30-3	8614.33	0.00	3.54	90-30-3	8611.08	0.00	3.22
90-30-4	8943.08	0.00	15.61	90-30-4	8911.15	0.00	5.60	90-30-4	8913.00	0.00	4.32
90-30-5	7957.45	0.00	23.08	90-30-5	7939.72	0.00	3.37	90-30-5	7943.25	0.00	0.88

Table 12 Results of Medium-Size Standard-Weight Instances in S_B with Granularity 1

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
90-30-1	9534.25	0.00	138.66	90-30-1	9512.66	0.00	121.06	90-30-1	9500.75	0.00	17.14
90-30-2	8520.95	0.00	18.59	90-30-2	8478.83	0.00	3.13	90-30-2	8472.31	0.00	5.98
90-30-3	8629.05	0.00	40.57	90-30-3	8613.61	0.00	4.04	90-30-3	8610.36	0.00	3.21
90-30-4	8942.18	0.00	14.10	90-30-4	8910.37	0.00	5.02	90-30-4	8912.49	0.00	3.22
90-30-5	7956.74	0.00	19.17	90-30-5	7939.18	0.00	3.69	90-30-5	7942.67	0.00	1.04

Computational Results for Instance Set S_B

2. Instances with 1-Second Time Granularity

2.1 Uniform weights

Table 13 Summary of the Results for the WTFRP on Uniform-Weight Instances with 1-Second Granularity in S_B

			Step 1			*Step* 2	*Step* 3			*Step* 4		
Name	Inst	Switch	LB‰	$\lvert\mathcal{T}^1\rvert$	$\lvert\mathcal{U}^1\rvert$	UB‰	$\lvert\mathcal{U}^2\rvert$	$\lvert\mathcal{U}^3\rvert$	$\lvert\mathcal{U}^4\rvert$	$\lvert\hat{\mathcal{T}}\rvert$	$\lvert\hat{\mathcal{U}}\rvert$	CPU
60-10-0.5	5	0	8.44	8.0	170.6	0.00	99.0	50.2	48.0	0.2	0.2	0.1
60-10-0.6	5	0	2.62	5.4	151.0	0.00	102.2	45.4	38.2	0.0	0.0	0.1
60-10-0.7	5	0	1.83	5.2	145.6	0.00	65.6	26.8	22.2	0.0	0.0	0.1
60-20-0.5	5	0	2.11	15.8	526.0	0.27	597.8	153.6	137.2	2.0	4.8	0.5
60-20-0.6	5	0	1.75	13.4	459.8	0.00	434.4	123.0	109.4	0.0	0.0	0.3
60-20-0.7	5	0	1.09	10.2	436.4	0.00	284.6	89.8	76.4	0.0	0.0	0.2
60-30-0.5	5	1	2.82	40.6	1768.6	0.98	15790.4	1814.4	1557.4	0.8	4.4	7.1
60-30-0.6	5	0	2.54	26.8	1077.8	0.12	4760.8	1112.0	944.0	0.8	5.8	2.5
60-30-0.7	5	0	0.62	20.8	1151.2	0.00	3405.4	358.2	324.6	0.0	0.0	0.7
90-10-0.5	4	0	2.14	11.8	363.0	0.00	282.0	71.0	64.5	0.0	0.0	0.1
90-10-0.6	5	0	3.31	10.4	332.0	0.22	408.0	157.4	146.2	0.0	0.0	0.3
90-10-0.7	5	0	3.08	6.8	306.2	0.00	450.0	213.2	187.8	0.0	0.0	0.1
90-20-0.5	5	1	2.56	30.8	2346.2	0.00	33250.2	2525.8	2034.2	1.6	57.0	7.2
90-20-0.6	5	0	1.29	23.6	1432.0	0.00	11593.8	1388.2	1126.6	0.2	0.0	1.1
90-20-0.7	5	1	0.33	20.6	1201.4	0.00	36452.6	802.2	664.4	0.0	0.0	2.3
90-30-0.5	5	3	1.96	43.6	7511.4	0.06	1259350.2	13091.2	10716.6	0.0	0.0	58.4
90-30-0.6	5	1	0.97	33.6	5037.6	0.00	414706.0	14431.8	10271.8	0.4	0.0	26.5
90-30-0.7	5	0	0.49	30.0	4541.2	0.00	39481.4	5026.4	4473.0	0.0	0.0	5.4
120-10-0.5	4	0	9.48	15.5	940.8	0.05	10381.5	5310.3	4689.0	0.3	41.0	2.8
120-10-0.6	5	0	5.56	10.8	879.4	0.02	7867.0	3597.0	3136.4	0.0	0.0	1.0
120-10-0.7	5	0	4.05	8.4	832.6	0.00	5444.6	2900.6	2705.6	0.0	0.0	0.5
120-20-0.5	5	2	5.77	45.6	5575.8	0.90	1054988.6	104319.8	72926.0	0.6	324.4	264.6
120-20-0.6	5	2	1.75	26.0	5921.0	0.02	878396.6	62244.8	55157.4	0.2	986.6	69.8
120-20-0.7	5	0	2.39	20.8	3571.2	0.02	193304.0	56078.6	50040.8	0.0	0.0	30.3
120-30-0.5	4(1)	3	1.83	59.5	10423.8	0.39	10663528.0	230059.8	210788.5	0.0	0.0	1862.3
120-30-0.6	5	2	0.66	45.2	8965.0	0.04	5125192.6	100056.0	94613.2	0.4	29.8	1079.6
120-30-0.7	5(1)	0	0.75	41.6	7631.2	0.06	1946668.4	103450.2	100091.4	0.0	0.0	396.5

Computational Results for Instance Set S_B

2. Instances with 1-Second Time Granularity

2.1 Uniform weights

Table 14 Summary of the Results for Instances with Uniform Weights and Granularity 1

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-1	1032.67	0.00	0.30	60-10-1	1029.77	0.00	0.10	60-10-1	1029.00	0.00	0.05
60-10-2	1056.38	0.00	0.16	60-10-2	1048.56	0.00	0.04	60-10-2	1048.76	0.00	0.13
60-10-3	1066.86	0.00	0.15	60-10-3	1064.52	0.00	0.14	60-10-3	1063.25	0.00	0.07
60-10-4	1054.53	0.00	0.05	60-10-4	1049.02	0.00	0.09	60-10-4	1049.57	0.00	0.07
60-10-5	1054.13	0.00	0.09	60-10-5	1052.43	0.00	0.07	60-10-5	1050.58	0.00	0.07
60-20-1	2133.33	0.00	0.75	60-20-1	2129.83	0.00	0.20	60-20-1	2127.61	0.00	0.08
60-20-2	2114.16	0.00	0.50	60-20-2	2109.92	0.00	0.37	60-20-2	2106.94	0.00	0.34
60-20-3	2012.97	0.00	0.22	60-20-3	2014.14	0.00	0.47	60-20-3	2010.36	0.00	0.13
60-20-4	2006.85	0.00	0.25	60-20-4	2002.35	0.00	0.17	60-20-4	2000.57	0.00	0.18
60-20-5	2136.04	0.00	0.61	60-20-5	2128.89	0.00	0.24	60-20-5	2127.53	0.00	0.09
60-30-1	3094.13	0.00	1.08	60-30-1	3089.58	0.00	0.39	60-30-1	3085.21	0.00	0.81
60-30-2	3174.59	0.00	3.40	60-30-2	3168.24	0.00	8.28	60-30-2	3157.69	0.00	0.91
60-30-3	3160.79	0.00	2.55	60-30-3	3151.37	0.00	1.05	60-30-3	3148.10	0.00	0.59
60-30-4	3163.59	0.00	25	60-30-4	3144.28	0.00	1.82	60-30-4	3142.35	0.00	0.60
60-30-5	3176.07	0.00	3.30	60-30-5	3157.75	0.00	0.83	60-30-5	3153.47	0.00	0.36
90-10-1	1616.12	0.00	0.11	90-10-1	1616.97	0.00	0.16	90-10-1	1615.43	0.00	0.13
90-10-2	1661.55	0.00	0.18	90-10-2	1657.71	0.00	0.22	90-10-2	1654.32	0.00	0.07
90-10-3	1612.52	0.00	0.05	90-10-3	1608.18	0.00	0.05	90-10-3	1605.91	0.00	0.05
90-10-4	–	–	–	90-10-4	1632.11	0.00	0.07	90-10-4	1633.22	0.00	0.11
90-10-5	1626.50	0.00	0.10	90-10-5	1625.14	0.00	0.83	90-10-5	1624.14	0.00	0.08
90-20-1	3274.82	0.00	0.49	90-20-1	3269.88	0.00	0.62	90-20-1	3265.72	0.00	0.81
90-20-2	3294.67	0.00	29.79	90-20-2	3286.19	0.00	1.63	90-20-2	3281.67	0.00	0.32
90-20-3	3293.48	0.00	1.45	90-20-3	3288.52	0.00	0.43	90-20-3	3286.51	0.00	0.37
90-20-4	3291.29	0.00	1.39	90-20-4	3283.53	0.00	2.09	90-20-4	3281.33	0.00	9.70
90-20-5	3263.17	0.00	2.75	90-20-5	3248.24	0.00	0.92	90-20-5	3244.59	0.00	0.27
90-30-1	4902.59	0.00	178.31	90-30-1	4889.46	0.00	74.12	90-30-1	4886.93	0.00	10.19
90-30-2	4803.78	0.00	34.45	90-30-2	4786.64	0.00	4.92	90-30-2	4780.06	0.00	4.37
90-30-3	4965.78	0.00	55.23	90-30-3	4956.97	0.00	4.16	90-30-3	4956.28	0.00	9.21
90-30-4	4675.14	0.00	15.25	90-30-4	4665.67	0.00	43.73	90-30-4	4662.03	0.00	1.71
90-30-5	4881.07	0.00	8.85	90-30-5	4873.20	0.00	5.70	90-30-5	4871.57	0.00	1.48
120-10-1	2195.90	0.00	7.41	120-10-1	2193.25	0.00	2.40	120-10-1	2191.21	0.00	1.21
120-10-2	2200.70	0.00	0.33	120-10-2	2200.21	0.00	0.26	120-10-2	2198.78	0.00	0.07
120-10-3	–	–	–	120-10-3	2201.95	0.00	1	120-10-3	2198.88	0.00	0.14
120-10-4	2151.52	0.00	2.73	120-10-4	2148.67	0.00	0.73	120-10-4	2149.36	0.00	0.84
120-10-5	2234.86	0.00	0.59	120-10-5	2229.97	0.00	0.42	120-10-5	2228.95	0.00	0.29
120-20-1	4529.27	0.00	35.88	120-20-1	4524.85	0.00	5.07	120-20-1	4524.73	0.00	3.37
120-20-2	4496.84	0.00	26.38	120-20-2	4489.66	0.00	4.95	120-20-2	4482.61	0.00	0.95
120-20-3	4474.83	0.00	439.23	120-20-3	4458.51	0.00	24.89	120-20-3	4455.80	0.00	106.97
120-20-4	4385.61	0.00	151.39	120-20-4	4384.31	0.00	281.76	120-20-4	4380.75	0.00	33.62
120-20-5	4459.03	0.00	670.13	120-20-5	4438.32	0.00	32.54	120-20-5	4438.05	0.00	6.81
120-30-1	6662.35	2.90	3600.00(T)	120-30-1	6642.35	0.00	1126.31	120-30-1	6639.42	0.00	11.36
120-30-2	6715.91	0.00	1095.92	120-30-2	6708.66	0.00	290.97	120-30-2	6708.89	0.00	146.80
120-30-3	6628.26	0.00	859.14	120-30-3	6619.48	0.00	714.68	120-30-3	6616.98	0.00	665.11
120-30-4	6644.99	0.00	1894.08	120-30-4	6626.59	0.00	3172.35	120-30-4	6621.88	1.98	1058.18(M)
120-30-5	–	–	–	120-30-5	6553.91	0.00	93.50	120-30-5	6549.08	0.00	100.87

Computational Results for Instance Set S_B

2. Instances with 1-Second Time Granularity

2.2 Standard weights

Table 15 Summary of the Results for the WTFRP on Standard-Weight Instances with 1-Second Granularity in S_B

			Step 1			*Step* 2	*Step* 3			*Step* 4		
Name	Inst	Switch	LB‰	$\lvert\mathcal{T}^1\rvert$	$\lvert\mathcal{U}^1\rvert$	UB‰	$\lvert\mathcal{U}^2\rvert$	$\lvert\mathcal{U}^3\rvert$	$\lvert\mathcal{U}^4\rvert$	$\lvert\hat{\mathcal{T}}\rvert$	$\lvert\hat{\mathcal{U}}\rvert$	CPU
60-10-0.5	5	0	2.22	8.0	118.6	0.00	69.2	27.2	24.8	0.0	0.0	0.1
60-10-0.6	5	0	0.71	5.6	137.6	0.00	55.0	19.4	16.4	0.0	0.0	0.0
60-10-0.7	5	0	0.93	5.8	115.6	0.00	85.6	32.0	25.6	0.0	0.0	0.0
60-20-0.5	5	2	3.63	18.2	407.4	0.92	3398.6	410.6	392.8	0.2	1.0	0.9
60-20-0.6	5	0	1.54	16.6	276.0	0.09	516.2	136.4	122.8	0.2	0.4	0.2
60-20-0.7	5	1	1.23	12.0	364.4	0.00	2364.4	134.8	120.8	0.0	0.0	0.2
60-30-0.5	5	2	5.26	38.6	938.6	1.10	28055.4	4427.4	4212.2	5.2	79.0	9.4
60-30-0.6	5	2	1.92	29.2	704.0	0.21	27567.2	1484.0	1426.0	0.6	50.0	4.8
60-30-0.7	5	1	0.81	23.4	548.0	0.63	8540.4	728.2	708.0	0.6	1.4	0.7
90-10-0.5	5	0	4.06	11.6	271.6	0.00	480.0	177.0	170.8	0.0	0.0	0.3
90-10-0.6	5	1	5.40	9.6	287.6	0.08	1429.2	281.6	266.4	0.0	0.0	0.7
90-10-0.7	5	0	5.14	6.8	234.0	0.03	513.0	280.6	274.2	0.4	0.2	0.1
90-20-0.5	5	0	1.97	31.8	977.8	0.17	10391.2	1168.4	1120.8	0.0	0.0	1.7
90-20-0.6	5	0	1.82	24.6	763.8	0.45	14598.4	2452.2	2191.8	0.0	0.0	1.7
90-20-0.7	5	1	0.33	18.4	737.6	0.01	31487.2	667.8	623.4	0.0	0.0	2.4
90-30-0.5	5	2	3.11	43.8	2883.8	0.52	899433.2	16069.0	14187.2	0.8	195.8	46.2
90-30-0.6	5	1	1.04	32.0	1908.6	0.00	662659.8	13131.2	10977.0	0.2	5.2	27.4
90-30-0.7	5	0	1.18	31.2	2376.4	0.00	85556.2	12388.8	11213.4	0.0	0.0	6.1
120-10-0.5	5	1	8.52	17.6	630.8	2.66	17778.6	4439.0	4124.0	0.4	5.0	2.9
120-10-0.6	5	0	6.73	12.2	523.0	0.00	6176.4	3473.4	3400.0	0.0	0.0	1.0
120-10-0.7	5	0	7	11.2	555.0	0.35	5562.8	3631.4	3593.4	0.0	0.0	1.1
120-20-0.5	5	2	6.43	45.8	2329.8	0.26	1903651.0	90525.2	82959.4	0.8	48.8	149.3
120-20-0.6	5	1	3.12	29.4	2228.2	0.23	560746.8	43610.6	41767.0	0.4	360.0	26.0
120-20-0.7	5	0	2.50	26.2	1580.4	0.07	203563.2	53741.8	52106.6	0.0	0.0	20.8
120-30-0.5	4(1)	1	2.23	58.8	8793.3	0.00	7419707.5	235912.0	217170.8	1.0	64.0	1750.6
120-30-0.6	5	0	0.67	41.6	7692.2	0.09	3539465.2	95743.8	87028.8	0.8	36.4	655.6
120-30-0.7	5(1)	0	1.02	42.0	8128.2	0.00	490823.6	83160.4	81834.0	0.0	0.0	277.4

Computational Results for Instance Set S_B

2. Instances with 1-Second Time Granularity

2.2 Standard weights

Table 16 Detailed Results of Standard-Weight Instances in S_B with 1-Second Time Granularity

ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-1	1859.23	0.00	0.24	60-10-1	1857.32	0.00	0.08	60-10-1	1855.66	0.00	0.06
60-10-2	1999.74	0.00	0.10	60-10-2	1991.84	0.00	0.02	60-10-2	1991.72	0.00	0.05
60-10-3	2260.34	0.00	0.06	60-10-3	2256.21	0.00	0.03	60-10-3	2255.07	0.00	0.02
60-10-4	2019.79	0.00	0.06	60-10-4	2004.37	0.00	0.03	60-10-4	2002.80	0.00	0.07
60-10-5	1900.38	0.00	0.03	60-10-5	1900.46	0.00	0.05	60-10-5	1897.32	0.00	0.05
60-20-1	3740.41	0.00	0.25	60-20-1	3734.14	0.00	0.31	60-20-1	3731.04	0.00	0.05
60-20-2	3921.11	0.00	2.21	60-20-2	3908.59	0.00	0.15	60-20-2	3901.47	0.00	0.61
60-20-3	3186.12	0.00	1.86	60-20-3	3175.15	0.00	0.20	60-20-3	3169.06	0.00	0.15
60-20-4	3806.88	0.00	0.17	60-20-4	3798.74	0.00	0.20	60-20-4	3796.96	0.00	0.07
60-20-5	3632.94	0.00	0.27	60-20-5	3621.38	0.00	0.12	60-20-5	3620.97	0.00	0.03
60-30-1	6645.22	0.00	2.04	60-30-1	6638.13	0.00	0.17	60-30-1	6628.88	0.00	0.85
60-30-2	5736.35	0.00	19.80	60-30-2	5704.14	0.00	17.42	60-30-2	5691.82	0.00	0.44
60-30-3	5298.99	0.00	12.44	60-30-3	5272.94	0.00	0.50	60-30-3	5266.68	0.00	0.50
60-30-4	5491.95	0.00	7.83	60-30-4	5467.38	0.00	1.94	60-30-4	5461.61	0.00	0.66
60-30-5	5296.34	0.00	5.12	60-30-5	5270.45	0.00	3.92	60-30-5	5258.89	0.00	1.22
90-10-1	2907.87	0.00	0.55	90-10-1	2906.42	0.00	0.25	90-10-1	2909.39	0.00	0.24
90-10-2	2995.87	0.00	0.45	90-10-2	2985.21	0.00	0.25	90-10-2	2981.07	0.00	0.10
90-10-3	2922.65	0.00	0.13	90-10-3	2911.74	0.00	0.07	90-10-3	2908.22	0.00	0.04
90-10-4	3156.39	0.00	0.14	90-10-4	3151.57	0.00	0.21	90-10-4	3150.84	0.00	0.17
90-10-5	2608.79	0.00	0.04	90-10-5	2606.61	0.00	2.93	90-10-5	2604.07	0.00	0.09
90-20-1	6507.74	0.00	0.99	90-20-1	6477.76	0.00	0.69	90-20-1	6474.81	0.00	1.16
90-20-2	6783.09	0.00	2.76	90-20-2	6771.15	0.00	2.17	90-20-2	6760.03	0.00	0.20
90-20-3	6108.01	0.00	1.20	90-20-3	6090.41	0.00	0.27	90-20-3	6091.03	0.00	0.20
90-20-4	5257.72	0.00	1.88	90-20-4	5246.83	0.00	4.88	90-20-4	5241.73	0.00	9.75
90-20-5	5980.39	0.00	1.60	90-20-5	5946.36	0.00	0.55	90-20-5	5945.04	0.00	0.50
90-30-1	9534.25	0.00	138.66	90-30-1	9512.66	0.00	121.06	90-30-1	9500.75	0.00	17.14
90-30-2	8520.95	0.00	18.59	90-30-2	8478.83	0.00	3.13	90-30-2	8472.31	0.00	5.98
90-30-3	8629.05	0.00	40.57	90-30-3	8613.61	0.00	4.04	90-30-3	8610.36	0.00	3.21
90-30-4	8942.18	0.00	14.10	90-30-4	8910.37	0.00	5.02	90-30-4	8912.49	0.00	3.22
90-30-5	7956.74	0.00	19.17	90-30-5	7939.18	0.00	3.69	90-30-5	7942.67	0.00	1.04
120-10-1	3861.84	0.00	2.65	120-10-1	3857.57	0.00	1.91	120-10-1	3854.95	0.00	3.40
120-10-2	3547.55	0.00	0.32	120-10-2	3542.99	0.00	0.05	120-10-2	3540.09	0.00	0.03
120-10-3	3419.65	0.00	0.73	120-10-3	3400.17	0.00	0.37	120-10-3	3397.93	0.00	0.09
120-10-4	4298.57	0.00	4.01	120-10-4	4293.41	0.00	2.29	120-10-4	4291.76	0.00	2.11
120-10-5	3825.76	0.00	7	120-10-5	3816.87	0.00	0.28	120-10-5	3808.00	0.00	0.11
120-20-1	8038.72	0.00	247.54	120-20-1	8028.39	0.00	4.17	120-20-1	8025.76	0.00	6.13
120-20-2	9270.73	0.00	31.95	120-20-2	9240.79	0.00	1.61	120-20-2	9233.91	0.00	0.37
120-20-3	9623.13	0.00	327.34	120-20-3	9589.35	0.00	35.48	120-20-3	9576.99	0.00	47.30
120-20-4	7707.35	0.00	123.01	120-20-4	7700.79	0.00	70.62	120-20-4	7697.83	0.00	41.81
120-20-5	7486.23	0.00	16.82	120-20-5	7447.75	0.00	17.87	120-20-5	7440.68	0.00	8.55
120-30-1	12970.10	2.75	3600.00(T)	120-30-1	12927.19	0.00	1247.68	120-30-1	12928.70	0.00	12.13
120-30-2	11870.88	0.00	119.87	120-30-2	11857.96	0.00	152.94	120-30-2	11853.76	0.00	9.28
120-30-3	11149.55	0.00	1854.13	120-30-3	11130.46	0.00	534.87	120-30-3	11126.50	0.00	7.64
120-30-4	11379.24	0.00	1428.60	120-30-4	11341.99	0.00	1121.19	120-30-4	11337.79	2.82	1088.28(M)
120-30-5	–	–	–	120-30-5	13266.76	0.00	221.49	120-30-5	13245.06	0.00	269.72